AzurESky "ensure quality of test"	**DEVELOPMENT AND MONITORING OF CONTROL CHARTS** **Document no:** **ASMF6.4.10 PROCEDURE**

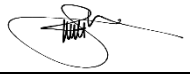
Compiled by:	**Sifiso Nyambi - Laboratory Manager**	
Approved by:	**Nonhlanhla Msibi - Operations Manager**	

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Purpose

To Define the procedure for creating, maintaining, and interpreting control charts to monitor the stability and performance of analytical processes and instruments in the laboratory, detect trends or shifts, and take corrective actions when results fall outside the control limits.

Application

This procedure applies to all laboratory measurements where **quality control (QC) samples**, **reference materials (RMs)**, or **proficiency test materials** are analyzed to assess equipment and method performance. It is applicable to both chemical and physical testing in the laboratory.

References

- ISO 17034:2016 – RM Production & QC Monitoring
- ISO 7870 – Control charts

Definitions

Control Chart: A statistical tool used to track measurement results over time and identify when a process is out of control.

Mean ($\bar{x}$): Average of the QC results used as the central line on the chart.

UCL/LCL: Upper and Lower Control Limits, usually set at a parameter's reproducibility.

Responsibilities

- **Analyst:** Record QC results promptly and accurately, initiates investigations into out-of-control situations.
- **Laboratory Supervisor:** Review control charts weekly, investigate out-of-control situations.

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- **Laboratory Manager:** Ensure procedure is followed and control charts are archived.

Procedure

Selection of Certified Reference Material

Quality control measures must be undertaken by a stable certified reference material representative of routine samples.

Certified Reference Material must be stored under controlled conditions to prevent degradation.

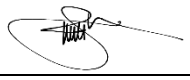
Development and evaluation of Control Charts

- When a new CRM is introduced into the system, part of its evaluation is the development of a control chart.
- This is done by analysing at least 10 samples
- Then conduct a statistical study of the runs to eliminate any outliers
- Find the mean of the sample which will become the laboratory mean value (i.e. The mean value must be reproducible to the provided mean value found on the CRM's certificate.)
- From the mean, the Upper Action Limit is found by adding the reproducibility Limit to the mean and for the Lower Action Limit, subtract the reproducibility limit from the mean.

  Mean Value ± Reproducibility Limit= Action Limits
- The Upper/Lower Warning Limit are calculated as follows:

  Mean Value ± (2/3 X reproducibility limit)
- Evaluate the Action limits in such a way that the upper action limit must be less than the maximum value provided in the certificate and the lower action limit must be more than the minimum value provided in the certificate.

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Day-to-Day Collection of Data

1. Analyse the CRM at defined intervals as outlined in each parameters' procedure.
2. Record the results on the control charts as and when a parameter was analysed on the day and connect the points with a line.
3. All results plotted on the control chart must be within the action limits.

Interpretation conditions

A control chart is said to be out of controlled limits if these conditions are detected:

- When three consecutive points fall outside either the Upper Warning Limit or the Lower Warning Limit.
- When there are 8 consecutive points either on the lower side of the mean/central line or on the upper side of the mean/central.
- When 8 consecutive points are gradually increasing or decreasing.
- "Spaced Bias" when results favour one side of the mean but break the bias limit once or twice in the process and still goes and favours the same side.

Remedial Actions

These are the routes to explore if a control chart deviates:

- When there is a noticeable trend that will lead to a deviating condition, it calls for an investigation into the instrument used. Act before the deviation occurs.
- This can be done by checking the furnace/oven verifications for any pattern that can be picked up and conducting the verification itself.
- In the case of Calorific value instruments, check if the oxygen is still above minimum and check for any leaks
- Thorough Maintenance of the bomb vessels and the calorimeters.
- Calibration is the last resort to any deviation picked up and should be conducted when all other options have been explored.
- In the process, open an IRQ and document the process for proof of investigation.

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- If the means to resolve the matter still falls short, take the matter to the reporting manager/supervisor.

Documentation

Plotting of points on Control charts are to be logged in this manner:

- Date, Analyst, original analysis, duplicate analysis, average result and plotting of the point on the correct spot.
- Control charts and IRQ's relating to the control charts are to be retained for two accreditation cycles (10 Years).

Revisions

Revision no	Content changed	Reason for revision

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